Yu-Hsiang Chen

Email ◇ GitHub ◇ LinkedIn ◇ Google Scholar ◇ Website

Education

National Yang Ming Chiao Tung University — Taiwan
M.Sc. in Computer Science — 09/2024 – 06/2026

National Tsing Hua University — Taiwan
B.Sc. in Technology Management — 09/2020 – 06/2024

- Ranked 2nd among 65 students every year. GPA 4.08/4.3.

Research Experience

University of California, Berkeley — Berkeley, CA
Researcher, MSC Lab; supervised by Prof. Masayoshi Tomizuka — 06/2025 – 12/2025

- Developed **ECoSim**, first data-efficient framework integrating diffusion and autoregressive models for controllable, realistic closed-loop traffic simulation; accepted at ECCV 2026.
- Created **HetroD**, the first high-fidelity drone dataset and benchmark for evaluating heterogeneous traffic in autonomous driving; accepted at ICRA 2026.

National Yang Ming Chiao Tung University — Hsinchu, Taiwan
Researcher, HCIS Lab; supervised by Prof. Yi-Ting Chen — 09/2024 – 06/2026

- Developed **NudgeVAD**, a language-guided end-to-end driving framework using FiLM residuals; won 1st place in the DriveX DoScenes Challenge at CVPR Workshops 2026.
- Co-developed frameworks for activation steering in end-to-end autonomous driving, safety validation, and VLM-based V2X scenario understanding, resulting in publications at IEEE ITSC 2026, ICCV Workshops 2025, and ECCV Workshops 2026.

National Tsing Hua University — Hsinchu, Taiwan
Undergraduate Researcher; supervised by Prof. Hung-Min Sun — 09/2022 – 06/2024

- Conducted NSTC-funded undergraduate research project on high-frequency trading; engineered a 400,000-record data pipeline and a campus application that improved operational efficiency by 45%.

Industry Experience

Synology — Taipei, Taiwan
Research Intern — 07/2026 – 02/2027

- Developed generalizable tiny VLM for fast, accurate open-vocabulary attribute prediction on resource-constrained NAS edge devices; under review at CVPR 2027.

Geniideas — Hsinchu, Taiwan
Research and Development Intern — 06/2023 – 06/2026

- Developed industrial AI systems for automated piping ISO generation, laser-welding inspection, and crystal-oscillator classification for clients including TSMC and Micron.

Selected Publications

ECoSim: Data-Efficient Fine-Tuning for Controllable Traffic Simulation — View
Y.-H. Chen , W.-J. Chang, Y.-T. Chen, and M. Tomizuka

- ECCV 2026.

Selected Publications (continued)

HetroD: A High-Fidelity Drone Dataset and Benchmark for Heterogeneous Traffic in Autonomous Driving View

Y.-H. Chen , W.-J. Chang, C. Kotulla, T. Keutgens, S. Runde, T. Moers, C. Klas, W. Zhan, M. Tomizuka, and Y.-T. Chen

- ICRA 2026.

NudgeVAD: Language-Nudged End-to-End Driving via FiLM Residuals View

C.-C. Yang*, **Y.-H. Chen** *, and Y.-T. Chen

- CVPR Workshops 2026 — 1st Place, DriveX DoScenes Challenge. *Co-first authors.

On the Feasibility of Activation Steering for End-to-End Driving View

S.-W. Lu, J.-S. Yoo, **Y.-H. Chen** , Y.-H. Tsai, W.-L. Chao, and Y.-T. Chen

- ECCV Workshops 2026.

V2X-Based Logical Scenario Understanding with Vision-Language Models View

C.-L. Chi, Z.-H. Li, **Y.-H. Chen** , and Y.-T. Chen

- ICCV Workshops 2025.

Academic Service

DriveX and X-Sense Workshops 2025 – 2026

Organizer, ICCV 2025 and ECCV 2026

- Launched the first Heterogeneous Traffic Simulation Challenge. View workshop

Conference and Journal Reviewer 2025 – Present

NeurIPS, CVPR, ECCV, ICRA, IEEE RA-L, IEEE Transactions on Reliability, and IJPRAI

Teaching Experience

National Yang Ming Chiao Tung University Taiwan

Teaching Assistant for Introduction to Artificial Intelligence and Physical AI 2025 – 2026

Awards and Scholarships

1st Place, DriveX DoScenes Challenge, CVPR Workshops 2026

Five-time Presidential Award for Academic Excellence, National Tsing Hua University 2020 – 2024

Outstanding Graduate Award, National Tsing Hua University 2024

Ministry of Education Scholarship for Overseas Internship, Taiwan 2024

Undergraduate Research Scholarship, National Science and Technology Council 2023

Outstanding Intern Award, Hsinchu County Government 2023

General Education Competition Award, National Tsing Hua University 2023

Technical Skills

- **Programming Languages:** Python, C++, C, JavaScript
- **ML/AI:** Autonomous Driving, Computer Vision, Generative AI, Mathematical Statistics, PyTorch